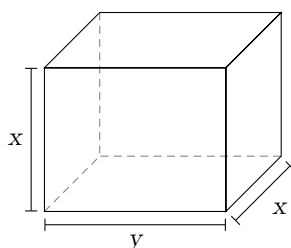


Problem 4

Problem 4

Problem 1 A box with two square sides is constructed to have a volume of 27. Denote the dimensions of the length of the base as y , and the width and height as x , as labeled in the diagram below. Find the dimensions needed to minimize the surface area of the box.



Solve the problem by performing the following steps.

- (a) Find a formula for $S(x)$ the surface area of the box, as a function of only the variable x .

Explanation. The volume of the box is given by $V = x^2y$ and the surface area is given by $S = 2x^2 + 4xy$. We are told that the volume of the box is 27, which means our constraint is that $x^2y = 27$, so that $y = \frac{27}{x^2}$. Plugging this into the surface area formula gives

$$S(x) = 2x^2 + 4x \left(\frac{27}{x^2} \right) = 2x^2 + \frac{108}{x}$$

- (b) Find the interval of interest for $S(x)$.

Explanation. Because x represents a side-length of this box, it cannot be negative. Moreover, the function $S(x)$ is not defined at $x = 0$, so x has to be positive. The Interval of Interest is $(0, \infty)$.

- (c) Find the x -value that gives the minimum surface area over all such boxes.
[2]

Explanation. Our objective function $S(x) = 2x^2 + \frac{108}{x}$ is continuous on the open interval $(0, \infty)$. This domain is an open interval, so to find the global minimum we start by finding the local extrema.

Problem 4

$S'(x) = 4x - \frac{108}{x^2}$. This exists for all nonzero values of x , and since $x = 0$ is not in our interval of interest, our critical points must have $S'(x) = 0$.

$$\begin{aligned} S'(x) &= 0 \\ 4x - \frac{108}{x^2} &= 0 \\ 4x &= \frac{108}{x^2} \\ x^3 &= 27 \\ x &= 3. \end{aligned}$$

There is one critical point, at $x = 3$. Since $S''(x) = 4 + \frac{216}{x^3}$, we have $S''(3) = 4 + \frac{216}{27} > 0$. By the Second Derivative Test, $x = 3$ is a local minimum.

Since $x = 3$ was the only local extremum on the open interval $(0, \infty)$, it is the global minimum as well.